



Greenhouse Gas (GHG) Emissions Analysis City of Calgary

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SAIT- Capstone Report

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1. Executive Summary

This case study presents an end-to-end data analytics project focused on analyzing greenhouse gas (GHG) emissions and energy consumption for the City of Calgary from 2000 to 2018. The objective was to transform fragmented environmental datasets into a structured, insight-driven business intelligence solution capable of supporting climate policy analysis and sustainability decision-making.

Through rigorous data cleaning, modeling, KPI design, and interactive dashboarding, the project delivers clear visibility into long-term emissions trends, sector-level drivers, and the relationship between energy use and GHG emissions.

2. Study Area

- **Primary focus:** City of Calgary
- **Benchmarking scope:** Canada-wide federal GHG datasets
- **GHG coverage:** CO₂ and other major greenhouse gases reported across sectors and years

3. Problem Statement

Governments at all levels face increasing pressure to meet climate commitments amid rising emissions and escalating carbon pricing policies. Despite the availability of emissions data, decision-makers often struggle with:

- Disconnected and inconsistent datasets
- Limited city-level analytical visibility
- Lack of standardized KPIs for emissions monitoring

This project addresses these challenges by consolidating federal and municipal datasets into a single analytical framework focused on the City of Calgary.

4. Project Objectives

- Consolidate multi-source GHG and energy datasets into an analytics-ready structure
- Clean and standardize emissions and energy data across years and sectors
- Design policy-relevant KPIs aligned with sustainability monitoring needs
- Build interactive dashboards to support trend analysis and benchmarking
- Translate analytical outputs into actionable insights

5. Data Sources Overview

The project utilized four Excel-based datasets covering GHG emissions and energy use from 2000–2018. These included cleaned emissions data, standardized energy consumption data, and supporting lookup tables for sector and category alignment.

Original datasets were provided as flat files with inconsistent units, naming conventions, and temporal granularity, requiring significant preprocessing before analysis.

6. Data Preparation & Cleaning

a. Initial Data Assessment

Each dataset was profiled to identify:

- Missing or null values

- Inconsistent units of measurement
- Duplicate or misaligned records
- Structural differences across years

b. Cleaning & Standardization Steps

- Converted all emissions and energy values into standardized units
- Harmonized sector and category naming across datasets
- Normalized year fields to support time-series analysis
- Removed duplicates and validated record-level consistency
- Performed reconciliation checks to ensure totals remained accurate after transformation

7. Data Modeling

Cleaned datasets were structured into an analytical model optimized for business intelligence:

- Fact tables capturing emissions and energy metrics
- Dimension tables for time, geography, sector, and energy source

This structure enabled flexible slicing, filtering, and aggregation within the visualization layer while maintaining data integrity.

8. KPIs & Analytics

Key performance indicators were designed to align with climate monitoring and policy analysis needs, including:

- Total GHG emissions by year
- Sector-wise emissions contribution
- Year-over-year emissions change
- Energy consumption by source and sector
- Comparative trend analysis across time

KPIs were implemented using calculated measures to ensure consistency across visuals.

9. Analytics & Visualization

Interactive dashboards were developed to enable:

- Exploration of long-term emissions trends (2000–2018)
- Identification of high-impact sectors
- Analysis of relationships between energy use and emissions
- Dynamic filtering by year, sector, and category

Visual design emphasizes clarity, executive readability, and analytical storytelling.

10. Key Findings

- GHG emissions exhibit distinct long-term patterns over the study period
- A limited number of sectors contribute a disproportionate share of total emissions
- Energy consumption patterns are strongly associated with emissions outcomes
- City-level trends highlight the importance of localized climate strategies rather than relying solely on national averages

11. Business & Policy Implications

The analytical outputs from this project support:

- Evidence-based climate policy discussions
- Identification of priority sectors for emissions reduction initiatives
- Improved transparency in municipal emissions reporting
- Ongoing monitoring of progress toward sustainability goals

12. Limitations

- Analysis limited to historical data up to 2018
- No predictive modeling included in the current scope
- Results depend on accuracy and completeness of reported source data

13. Future Enhancements

- Integrate emissions forecasting and scenario modeling
- Add per-capita and intensity-based KPIs
- Expand analysis to additional Canadian cities
- Incorporate policy simulation (e.g., carbon tax impact scenarios)

14. Conclusion

This case study demonstrates how structured data analytics and business intelligence can convert complex environmental datasets into actionable insights. By bridging raw data, analytical modeling, and visualization, the project highlights the critical role of analytics in supporting climate-focused decision-making at the municipal level.